THE DEGREE-ONE RELATION OF A CARTESIAN SIMPLICIAL SCHEME IS AN EQUIVALENCE RELATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the two 'Details omitted.' identifications (right-hand sides isomorphic to triple fibre products) are auxiliary rewirings with the reason named; the relation axioms themselves are all argued.

Editorial note. The label lemma-equivalence-relation names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Simplicial identities. Simplicial, Remark simplicial-remark-relations. Makes d_0 d_2 = d_1 d_0 and d_1 d_2 = d_1 d_1 meaningful for the displayed isomorphisms.
- Equivalence relation definition. Groupoids, Definition groupoids-definition-equivalence-relation. The target notion (monomorphism j, transitivity, reflexivity, symmetry) verified in the proof.

Source reference. Original reference 0249: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 6324–6324. Prerequisite pointer invoked at proof line(s) 6802; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 024A: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 6426–6426. Prerequisite pointer invoked at proof line(s) 6803; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 022P: sources/stacks-chow/tags.txt, source lines 2714–2714. Prerequisite pointer invoked at proof line(s) 6809; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original remark 016C: sources/stacks-chow/tags.txt, source lines 1539–1539. Prerequisite pointer invoked at proof line(s) 6823; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $\pi : Y \to (X/S)_\bullet$ be a cartesian morphism of simplicial schemes, see Definitions 0249 and 024A. Then the morphism*

$$j = (d_1^1, d_0^1) : Y_1 \to Y_0 \times_S Y_0$$

defines an equivalence relation on Y_0 over S, see Groupoids, Definition 022P.

Proof. Note that j is a monomorphism. Namely the composition $Y_1 \to Y_0 \times_S Y_0 \to Y_0 \times_S X$ is an isomorphism as π is cartesian.

Consider the morphism

$$(d_2^2, d_0^2) : Y_2 \to Y_1 \times_{d_0^1, Y_0, d_1^1} Y_1.$$

This works because $d_0 \circ d_2 = d_1 \circ d_0$, see Simplicial, Remark 016C. Also, it is a morphism over $(X/S)_2$. It is an isomorphism because $Y \to (X/S)_\bullet$ is cartesian. Note for example that the right hand side is isomorphic to $Y_0 \times_{\pi_0, X, \mathrm{pr}_1} (X \times_S X \times_S X) = X \times_S Y_0 \times_S X$ because π is cartesian. Details omitted.

As in Groupoids, Definition 022P we denote $t = \mathrm{pr}_0 \circ j = d_1^1$ and $s = \mathrm{pr}_1 \circ j = d_0^1$. The isomorphism above, combined with the morphism $d_1^2 : Y_2 \to Y_1$ give us a composition morphism

$$c : Y_1 \times_{s, Y_0, t} Y_1 \longrightarrow Y_1$$

over $Y_0 \times_S Y_0$. This immediately implies that for any scheme T/S the relation $Y_1(T) \subset Y_0(T) \times Y_0(T)$ is transitive.

Reflexivity follows from the fact that the restriction of the morphism j to the diagonal $\Delta : X \to X \times_S X$ is an isomorphism (again use the cartesian property of π).

To see symmetry we consider the morphism

$$(d_2^2, d_1^2) : Y_2 \to Y_1 \times_{d_1^1, Y_0, d_1^1} Y_1.$$

This works because $d_1 \circ d_2 = d_1 \circ d_1$, see Simplicial, Remark 016C. It is an isomorphism because $Y \to (X/S)_\bullet$ is cartesian. Note for example that the right hand side is isomorphic to $Y_0 \times_{\pi_0, X, \mathrm{pr}_0} (X \times_S X \times_S X) = Y_0 \times_S X \times_S X$ because π is cartesian. Details omitted.

Let T/S be a scheme. Let $a \sim b$ for $a, b \in Y_0(T)$ be synonymous with $(a, b) \in Y_1(T)$. The isomorphism (d_2^2, d_1^2) above implies that if $a \sim b$ and $a \sim c$, then $b \sim c$. Combined with reflexivity this shows that $\sim$ is an equivalence relation. $\square$